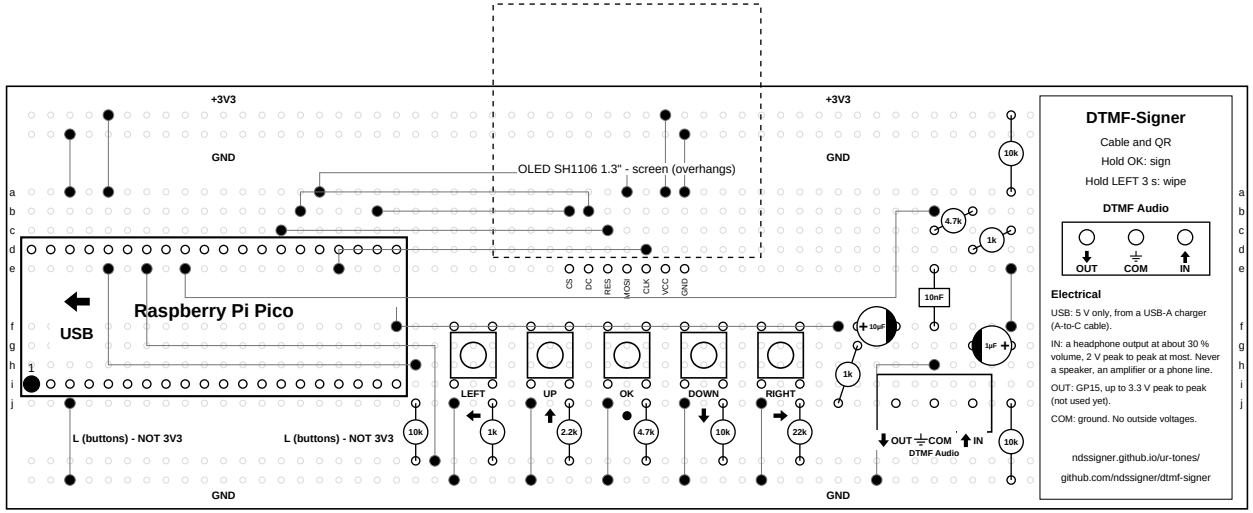
DTMF-Signer - Cable and QR - breadboard template

https://github.com/ndssigner/dtmf-signer · Hardware: CERN-OHL-S v2

ur-tones web tool
https://ndssigner.github.io/ur-tones/
Not on the board: it would confuse a wallet's camera.



Parts

1	Raspberry Pi Pico (RP2040), or a clone with its layout
1	1.3" SH1106 OLED, 128 x 64, SPI (7 pins)
5	Push button, 6 x 6 mm
3	Resistor 1 kΩ, 1/4 W
1	Resistor 2.2 kΩ, 1/4 W
2	Resistor 4.7 kΩ, 1/4 W
4	Resistor 10 kΩ, 1/4 W
1	Resistor 22 kΩ, 1/4 W
1	Electrolytic capacitor 1 µF
1	Electrolytic capacitor 10 µF
1	Film capacitor 10 nF
1	Screw terminal block, 3 ways, 5.08 mm
1	Full-size breadboard, or a solderable one
2 x 20, 1 x 7	Female header strips (sockets for the Pico and the screen)
	Solid insulated wire

Also: a USB cable and a USB-A charger, a wired headset with a microphone (to cut)

This bar must measure 100 mm.

PRINT AT ACTUAL SIZE (100 %)

Turn off "fit to page" or "scale to fit".
A4 landscape; on Letter paper it also fits at actual size.
Check the bar at the bottom: it must measure 100 mm.